rapidmesh

A Hierarchical Bottom-Up Centroidal-Voronoi Meshing Algorithm
for B-Rep Solids with Analytic and NURBS Geometry

Specification and mathematical derivation

June 17, 2026

Abstract

This document specifies the meshing algorithm rapidmesh uses to turn a boundary representation (B-Rep) of one or more material regions into a conforming, isotropic, high-quality tetrahedral mesh. The algorithm is *dimensionally hierarchical*: it meshes 0-cells (corners), then 1-cells (edges), then 2-cells (faces), then the 3-cell (volume), and *freezes* each level before the next consumes it. Within every dimension it combines *error-driven adaptive sampling* (insert where a geometric tolerance is violated) with *sizing-weighted Centroidal Voronoi Tessellation* (Lloyd relaxation) for point quality. The frozen surface triangulation is a *hard constraint* on the volume Delaunay, which is what makes the boundary watertight by construction. We derive, with a computer-algebra system, the arc-length, curvature, deflection-sizing, first/second fundamental form and distance-faithful chart formulae for every edge and face modality the geometry kernel supports (line, circle/arc, surface–surface intersection, NURBS curve, extruded profile; plane, cylinder, cone, sphere, torus, extruded/ruled, NURBS), and give pseudocode for each stage.

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1 Design philosophy

Principle 1.1 (Dimensional hierarchy)**.** *A k-cell is meshed only after every $(k-1)$-cell on its boundary is meshed and* frozen*. Corners pin edges; edges pin faces; faces pin the volume. Shared lower-dimensional entities are meshed once and reused, so adjacent cells agree on their common boundary exactly (conformity across seams is inherited, not reconstructed).*

Principle 1.2 (Two orthogonal mechanisms per level)**.** *Each dimension uses (i)* error-driven adaptivity *to decide* how many *points and* where *(insert at the worst geometric-error location, then re-relax, until a tolerance is met), and (ii)* CVT/Lloyd relaxation *to decide* point quality *(move each free site to the density-weighted centroid of its (restricted) Voronoi cell). Adaptivity controls density; CVT controls regularity.*

Principle 1.3 (Constrained volume, not recovered boundary). *The frozen surface triangulation enters the volume stage as a set of* constraint facets*. The volume tetrahedralization is required to contain them as faces. The boundary is therefore* enforced*, not statistically recovered: there are no straddling tetrahedra, hence no "noses" protruding into a void and no gaps. This is the decisive structural property and the reason the surface must be completed* before *the volume is seeded.*

Principle 1.4 (Embarrassingly parallel within each level). *Freezing a level before the next consumes it makes the entities* within *a level mutually independent: they share only already-frozen, read-only lower-dimensional boundary. Hence each level is processed in parallel with no synchronisation:*

- ***all edges*** *at once (each depends only on its two fixed corners);*
- *then* ***all faces*** *at once (each depends only on its fixed boundary edges, which are computed once and read concurrently);*
- *then* ***all volume regions*** *at once (each material region's constrained tetrahedralization depends only on its fixed bounding surface; shared interface facets are read-only).*

Shared lower-dimensional entities are meshed exactly once and cached, which is what removes write contention. Each entity's result is independently deterministic and merged in fixed id order, so parallelism does not affect reproducibility (section 2). The three barriers (edges $\rightarrow$ faces $\rightarrow$ volume) are the only synchronisation points.

The geometry source is a B-Rep assembled from the exact CSG arrangement: faces are trimmed analytic (or NURBS) surfaces, edges are analytic curves (including the intersection curves booleans create), vertices are corner points, and the radial-edge topology records which faces meet along each edge and the material region on each side. The mesh is built *from this geometry*, independent of any input tessellation. Figure 1 shows the whole pipeline.

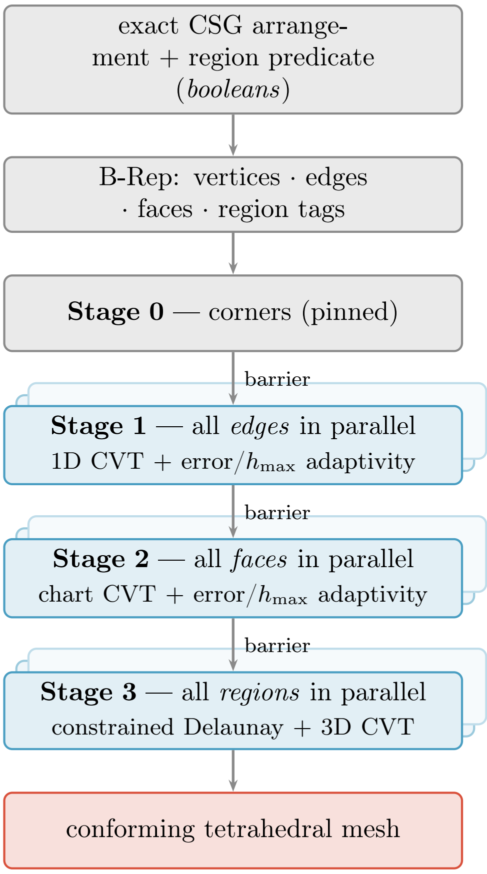

Figure 1: The bottom-up pipeline. Exact booleans produce the B-Rep; the mesher then meshes 0-, 1-, 2-, 3-cells in order. Within each of stages 1–3 the entities are independent and run in parallel (the stacked boxes); the only synchronisation is the barrier between levels (Principle 1.4).

2 Notation and preliminaries

2.1 The sizing field

A scalar *sizing field* $h : \mathbb{R}^3 \to \mathbb{R}_{>0}$ prescribes the target element edge length. It is assembled bottom-up (edges set it on 1-cells, faces refine it on 2-cells, the volume smooths it through the interior) and is always made *Lipschitz* (gradation-limited):

$$h(\mathbf{x}) \;\le\; h(\mathbf{y}) + \alpha\,\|\mathbf{x}-\mathbf{y}\| \qquad \forall\,\mathbf{x},\mathbf{y}, \tag{2.1}$$

with grading constant $\alpha \in (0,1]$ (the default $\alpha = \frac{1}{2}$ grows neighbouring elements by at most $\sim 1.5\times$). Equation (2.1) is the standard mesh-gradation constraint [1]. Given raw (ungraded) size requests $h_0(\mathbf{y})$ from the geometry, the largest field that satisfies both $h \le h_0$ and (2.1) is the lower envelope

$$h(\mathbf{x}) \;=\; \min_{\mathbf{y}} \big(h_0(\mathbf{y}) + \alpha\,\|\mathbf{x}-\mathbf{y}\|\big), \tag{2.2}$$

the viscosity solution of the Eikonal inequality $\|\nabla h\| \le \alpha$ with obstacle $h \le h_0$. It is computed by a single fast-marching (or a few Gauss–Seidel sweeps) over the size sources, $O(N \log N)$ in the number of grid nodes – the same machinery at every stage, fed first by edges, then faces, then the volume.

2.2 Centroidal Voronoi Tessellation

Given a density $\rho : \Omega \to \mathbb{R}_{>0}$ on a domain Ω and sites $\{\mathbf{z}_i\}_{i=1}^n$, the Voronoi region of $\mathbf{z}_i$ is $V_i = \{\mathbf{x} \in \Omega : \|\mathbf{x}-\mathbf{z}_i\| \le \|\mathbf{x}-\mathbf{z}_j\| \;\; \forall j\}$. The tessellation is *centroidal* when each site coincides with the mass centroid of its region,

$$\mathbf{z}_i \;=\; \mathbf{c}_i \;:=\; \frac{\displaystyle\int_{V_i} \rho(\mathbf{x})\,\mathbf{x}\,\mathrm{d}\mathbf{x}}{\displaystyle\int_{V_i} \rho(\mathbf{x})\,\mathrm{d}\mathbf{x}}. \tag{2.3}$$

A CVT is a critical point of the quantization energy

$$\mathcal{F}(\{\mathbf{z}_i\}) = \sum_i \int_{V_i} \rho(\mathbf{x})\,\|\mathbf{x}-\mathbf{z}_i\|^2\,\mathrm{d}\mathbf{x} \tag{2.4}$$

[2]. Holding the cells fixed and differentiating with respect to a single site,

$$\frac{\partial \mathcal{F}}{\partial \mathbf{z}_i} = -2\int_{V_i} \rho(\mathbf{x})\,(\mathbf{x}-\mathbf{z}_i)\,\mathrm{d}\mathbf{x}, \tag{2.5}$$

where the boundary terms from moving ∂V_i cancel because the integrand $\rho\,\|\mathbf{x}-\mathbf{z}_i\|^2$ is continuous across the shared Voronoi faces (each face contributes equal and opposite to its two cells). Setting (2.5) to zero gives $\mathbf{z}_i = \int_{V_i}\rho\mathbf{x}/\int_{V_i}\rho = \mathbf{c}_i$ – exactly the centroid (2.3). The one-dimensional case is immediate: $\partial_z\int_a^b(x-z)^2\,\mathrm{d}x = 0 \Rightarrow z = \frac{a+b}{2}$. *Lloyd's algorithm* [3] alternates "recompute V_i" and "move $\mathbf{z}_i \leftarrow \mathbf{c}_i$" and converges to a CVT; the resulting point set is blue-noise / isotropic, which is precisely the input a Delaunay mesher needs for well-shaped elements.

Density that realises a sizing field. To make the local point spacing follow h, choose

$$\rho(\mathbf{x}) \;=\; h(\mathbf{x})^{-(d+2)} \quad \text{(theory, } d \text{ dimensions)}, \qquad \rho(\mathbf{x}) \;=\; h(\mathbf{x})^{-d} \text{ (used in practice)}, \tag{2.6}$$

because a CVT equidistributes the energy density, giving local spacing $\propto \rho^{-1/(d+2)}$; the simpler h^{-d} (number density $\propto$ points per unit volume) is robust and is what we weight cells by. See [4], [5] for the exponent discussion.

2.3 Discrete centroids and the restricted setting

On a triangulation, the integral (2.3) is approximated by summing over the simplices incident to $\mathbf{z}_i$, each weighted by its measure times the density at its centroid. In dimension d with incident simplices T of centroid $\mathbf{g}_T$, measure $|T|$,

$$\mathbf{c}_i \;\approx\; \frac{\sum_{T\ni\mathbf{z}_i} |T|\;\rho(\mathbf{g}_T)\,\mathbf{g}_T}{\sum_{T\ni\mathbf{z}_i} |T|\;\rho(\mathbf{g}_T)}. \tag{2.7}$$

On a curved surface, the relevant object is the *restricted* Voronoi diagram (Voronoi cells clipped to the surface) and its dual *restricted Delaunay* triangulation [6], [7]; in practice we relax in a 2D chart (section 5) so (2.7) applies in the chart and the move is lifted back onto the surface.

2.4 Robust predicates

All combinatorial decisions (orientation, in-circle/in-sphere) use exact adaptive-precision predicates [8]; only non-decision quantities (centroid weights, sizing values) are floating point. Map/iteration order is made deterministic (seedless hashing), so the mesh is reproducible.

3 Stage 0 – Vertices

Every B-Rep corner $\mathbf{p}$ becomes a *pinned* site: it never moves and is present in every downstream triangulation. Pinning corners preserves sharp features exactly. Formally these are the 0-cells of the cell complex and the fixed points of all relaxations below.

4 Stage 1 – Edges

A B-Rep edge is a curve $\mathbf{C} : [t_0, t_1] \to \mathbb{R}^3$ between two corners. We support the modalities in table 1.

Table 1: Edge curve modalities. All expose $\mathbf{C}(t), \mathbf{C}'(t), \mathbf{C}''(t)$.

Modality	$\mathbf{C}(t)$	Source
Line	$\mathbf{p}_0 + t\,\mathbf{d}$	straight B-Rep edge
Circle / arc	$\mathbf{c} + r(\cos t\,\mathbf{x} + \sin t\,\mathbf{y})$	rim, fillet, boolean of quadrics
Intersection	implicit $\mathbf{S}_a \cap \mathbf{S}_b$, seeded by the chain	generic surface–surface boolean
NURBS	$\dfrac{\sum_i N_{i,p}(t)\,w_i\,\mathbf{P}_i}{\sum_i N_{i,p}(t)\,w_i}$	imported / profile curve [9]
Extruded profile	$\mathbf{b} + z\,\mathbf{a} + u_x\,P_x(t) + u_y\,P_y(t)$	lifted 2D profile (airfoil)
Polyline	piecewise linear chain	faceted fallback

4.1 Arc length

The arc length and the natural (unit-speed) parameter are

$$s(t) = \int_{t_0}^{t} \left\|\mathbf{C}'(\tau)\right\|\,\mathrm{d}\tau, \qquad \mathrm{d}s = \left\|\mathbf{C}'(t)\right\|\,\mathrm{d}t. \tag{4.1}$$

We tabulate $s(t)$ once (adaptive Gauss/Simpson) and invert $t(s)$ by table lookup + a Newton step using $\mathrm{d}t/\,\mathrm{d}s = 1/\,\|\mathbf{C}'\|$, so points can be placed at prescribed arc-length spacings. For a circle $\|\mathbf{C}'\| = r$ and $s = r\,\Delta t$ exactly; for a line it is exact; for NURBS / extruded profiles the table is used.

4.2 Curvature

For a space curve the (unsigned) curvature is

$$\kappa(t) \;=\; \frac{\|\mathbf{C}'(t) \times \mathbf{C}''(t)\|}{\|\mathbf{C}'(t)\|^3}, \qquad R(t) = \frac{1}{\kappa(t)} \text{ (radius of curvature)}, \tag{4.2}$$

derived from the Frenet apparatus and verified symbolically. A straight segment has $\kappa \equiv 0$, $R \equiv \infty$; a circle of radius r has $\kappa \equiv 1/r$. For a NURBS curve $\mathbf{C} = \mathbf{A}/w$ (homogeneous numerator $\mathbf{A} = \sum_i N_{i,p} w_i \mathbf{P}_i$, weight $w = \sum_i N_{i,p} w_i$) the derivatives follow from the quotient rule,

$$\mathbf{C}' = \frac{\mathbf{A}'w - \mathbf{A}w'}{w^2}, \qquad \mathbf{C}'' = \frac{\mathbf{A}'' - 2\mathbf{C}'w' - \mathbf{C}w''}{w}, \tag{4.3}$$

with $\mathbf{A}^{(k)}, w^{(k)}$ from the de Boor recurrence [9]; (4.1)–(4.2) then apply unchanged.

4.3 Intersection curves

A boolean between two surfaces $\mathbf{S}_a, \mathbf{S}_b$ creates an edge along their intersection. Writing the surfaces locally as implicit $f(\mathbf{x}) = 0$, $g(\mathbf{x}) = 0$ (signed distance to the analytic surface, so ∇f is the unit normal $\mathbf{n}_a$), the curve $\{f = g = 0\}$ has unit tangent

$$\mathbf{T} \;=\; \frac{\nabla f \times \nabla g}{\|\nabla f \times \nabla g\|} \;=\; \frac{\mathbf{n}_a \times \mathbf{n}_b}{\|\mathbf{n}_a \times \mathbf{n}_b\|}, \tag{4.4}$$

which degenerates only where the surfaces are tangent ($\mathbf{n}_a \parallel \mathbf{n}_b$). A point near the curve is projected *onto both surfaces* simultaneously by a constrained Newton iteration on $\mathbf{F} = (f, g)^\mathsf{T}$ with Jacobian $J = [\nabla f^\mathsf{T}; \nabla g^\mathsf{T}] \in \mathbb{R}^{2\times 3}$; the minimum-norm step is

$$\mathbf{x} \leftarrow \mathbf{x} - J^+ \mathbf{F}(\mathbf{x}), \qquad J^+ = J^\mathsf{T}\big(JJ^\mathsf{T}\big)^{-1} \ \ (2\times 2 \text{ inverse}), \tag{4.5}$$

quadratically convergent off the tangency locus. The B-rep chain seeds the iteration, and arc length / sizing then use the recovered $\mathbf{T}$ and the curve curvature

$$\kappa = \frac{\|\mathbf{C}' \times \mathbf{C}''\|}{\|\mathbf{C}'\|^3}, \qquad \mathbf{C}' \parallel \mathbf{T}, \quad \mathbf{C}'' = (\text{rate of change of } \mathbf{T} \text{ along the curve}), \tag{4.6}$$

obtained by differentiating (4.4) along $\mathbf{T}$ (or, where both surfaces are analytic with known second fundamental forms, from their normal curvatures in the direction $\mathbf{T}$). A circle is recovered exactly when both operands are quadrics sharing an axis, in which case the modality collapses to the *Circle* row of table 1.

4.4 Deflection sizing

The chord between two samples a parametric distance apart deviates from the true arc by the *sagitta.* For an arc of radius R subtending a half-angle $\theta/2$, with chord length $\ell = 2R\sin(\theta/2)$, the maximum chord–arc deviation is

$$\varepsilon = R\big(1 - \cos\tfrac{\theta}{2}\big) = \frac{R\,\theta^2}{8} + O(\theta^4) \approx \frac{\ell^2}{8R}. \tag{4.7}$$

Solving the small-angle relation for the admissible edge length given a *deflection tolerance* ε gives the local target

$$\boxed{\ell(\mathbf{x}) = \sqrt{8\,\varepsilon\, R(\mathbf{x})}} \qquad \Big(\text{exact: } \ell = 2\sqrt{R^2 - (R-\varepsilon)^2}\Big), \tag{4.8}$$

both verified with a computer-algebra system. The edge sizing field is then

$$h_{\text{edge}}(\mathbf{x}) = \min\Big(h_{\max},\ \sqrt{8\,\varepsilon\, R(\mathbf{x})}\Big), \tag{4.9}$$

i.e. refine where the curve bends (R small), coast at $h_{\max}$ where it is straight. This is the curve analogue of the surface bound (5.3).

4.5 1D CVT and error-driven adaptivity

On a fixed edge the n interior sites $s_1 < \cdots < s_n$ (arc-length coordinates, endpoints pinned) relax by the 1D centroid rule (the segment analogue of (2.7)): with neighbours s_{i-1}, s_{i+1} and density $\rho = h^{-1}$,

$$s_i \leftarrow \frac{\rho_{i-\frac12}\,\frac{s_{i-1}+s_i}{2} + \rho_{i+\frac12}\,\frac{s_i+s_{i+1}}{2}}{\rho_{i-\frac12} + \rho_{i+\frac12}}, \qquad \rho_{i\pm\frac12} = h\big(\tfrac{s_i+s_{i\pm1}}{2}\big)^{-1}, \tag{4.10}$$

which equidistributes spacing $\propto h$. Adaptivity wraps the relaxation (algorithm 1): relax for quality, then split wherever the chord deflection exceeds ε *or* a segment exceeds $h_{\max}$. All three meshing stages share this single relax–insert loop (figure 2); only the error measure and the centroid rule change with the dimension.

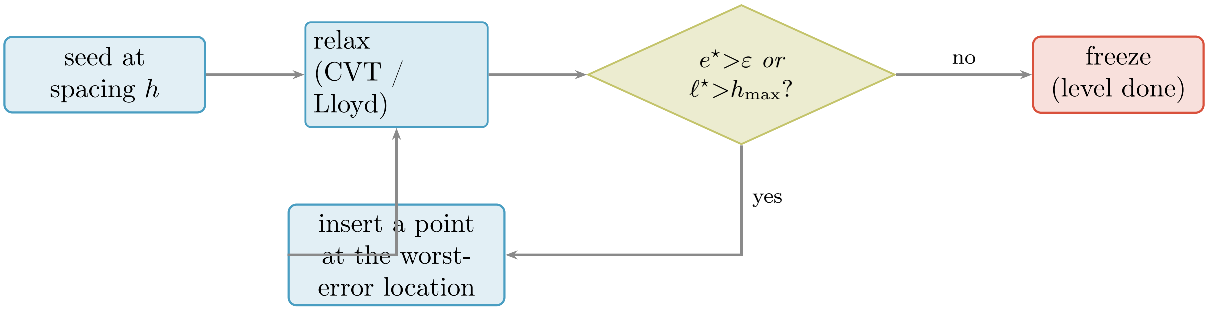

Figure 2: The relax–insert–relax scheme shared by all stages (Principle 1.4, mechanism (i)+(ii)). The error measure $e^\star$ is the chord deflection (edges), the triangle–surface deflection (faces) or the element quality (volume); $\ell^\star$ is the longest edge for the $h_{\max}$ bound. Iterate until both the tolerance and $h_{\max}$ hold and the sites stop moving, then freeze.

Algorithm 1: MESHEDGE($\mathbf{C}, h, \varepsilon$)

Input: edge curve $\mathbf{C}$; sizing field h; deflection tolerance ε
Output: frozen edge points (corners included)
tabulate $s(t)$; $L \leftarrow s(t_1)$;
$n \leftarrow \max\big(1,\ \lceil \int_0^L h(s)^{-1}\,\mathrm{d}s \rceil\big)$; `// initial count from the sizing field`
place $s_1, \ldots, s_n$ equidistributing h (cumulative-ρ inversion);
repeat
  for *a few iterations* **do**
    relax all s_i by (4.10);
  $e^\star, s^\star \leftarrow$ worst chord–arc deflection and its location ; `// error-driven`
  $\ell^\star, m^\star \leftarrow$ longest segment length and its midpoint ; `//` $h_{\max}$`-driven`
  if $e^\star > \varepsilon$ ***or*** $\ell^\star > h_{\max}$ **then**
    insert a site (at $s^\star$ if $e^\star>\varepsilon$, else $m^\star$); $n \mathrel{+}= 1$;
until $e^\star \le \varepsilon$ ***and*** $\ell^\star \le h_{\max}$ ***and*** *max site move* $< \tau_{\text{move}}$;
return $\{\mathbf{C}(s_i)\}$;

Shared edges are meshed once and cached, so both incident faces read identical points – and all edges run in parallel (Principle 1.4).

5 Stage 2 – Faces

A face is a trimmed surface $\mathbf{S} : \Omega \subset \mathbb{R}^2 \to \mathbb{R}^3$, $(u, v) \mapsto \mathbf{S}(u, v)$, bounded by oriented loops of frozen edges. The strategy is: parameterise the face by a *distance-faithful chart*, relax interior points there with a 2D surface-CVT (boundary = frozen edge points, held fixed), insert where the surface deflection tolerance is violated, and lift every chart point exactly onto $\mathbf{S}$. The output is a *conforming triangulation of the face* that the volume stage will treat as a hard constraint.

Face categories and the face sizing field. Three chart families cover all modalities (section 5.3): *planar* faces use the plane chart; *developable* analytic faces (cylinder, cone, extruded) use an exact isometric unroll; *doubly-curved* analytic faces (sphere, torus) and *periodic* full revolutions use a distance-faithful chart or the periodic revolution construction, with a restricted-CVT fallback for general NURBS. Before any face is seeded, its sizing field is assembled from the *now-frozen edge sampling* on its boundary (the edge spacings and any per-region $h_{\max}$) combined with the surface deflection bound (5.3), then Lipschitz-smoothed (2.1) across the face. The frozen edges therefore both bound the face and seed its density.

5.1 First fundamental form and area

With $\mathbf{S}_u = \partial\mathbf{S}/\partial u$, $\mathbf{S}_v = \partial\mathbf{S}/\partial v$, the metric (first fundamental form) coefficients and area element are

$$E = \mathbf{S}_u\cdot\mathbf{S}_u, \quad F = \mathbf{S}_u\cdot\mathbf{S}_v, \quad G = \mathbf{S}_v\cdot\mathbf{S}_v, \qquad \mathrm{d}A = \sqrt{EG - F^2}\;\mathrm{d}u\,\mathrm{d}v. \tag{5.1}$$

A curve $t \mapsto (u(t), v(t))$ in parameter space has length $\int \sqrt{E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2}\,\mathrm{d}t$; this is the metric the chart must reproduce for Euclidean 2D operations (separation, centroid) to mean arc length on the surface. Table 2 lists E, F, G, $\mathrm{d}A$ for every analytic modality, all derived with a computer-algebra system.

Worked example (sphere). For $\mathbf{S}(\lambda, \varphi) = r(\cos\varphi\cos\lambda, \cos\varphi\sin\lambda, \sin\varphi)$,

$$\mathbf{S}_u = r(-\cos\varphi\sin\lambda,\ \cos\varphi\cos\lambda,\ 0), \qquad \mathbf{S}_v = r(-\sin\varphi\cos\lambda,\ -\sin\varphi\sin\lambda,\ \cos\varphi),$$

so $E = \mathbf{S}_u\cdot\mathbf{S}_u = r^2\cos^2\varphi$, $F = \mathbf{S}_u\cdot\mathbf{S}_v = 0$, $G = \mathbf{S}_v\cdot\mathbf{S}_v = r^2$, giving $\mathrm{d}A = r^2\,|\cos\varphi|\,\mathrm{d}\lambda\,\mathrm{d}\varphi$ (the $\cos\varphi$ is the polar foreshortening). With outward normal $\mathbf{n} = \mathbf{S}/r$ the second fundamental form is $L = \mathbf{S}_{\lambda\lambda}\cdot\mathbf{n} = -r\cos^2\varphi$, $M = \mathbf{S}_{\lambda\varphi}\cdot\mathbf{n} = 0$, $N = \mathbf{S}_{\varphi\varphi}\cdot\mathbf{n} = -r$, hence the principal curvatures $\kappa_1 = L/E = -1/r$ and $\kappa_2 = N/G = -1/r$ are equal (the sphere is umbilic) and $K = \kappa_1\kappa_2 = 1/r^2$ – the table 2 sphere row. The other rows follow the same computation.

5.2 Curvature and surface deflection sizing

The shape operator's eigenvalues are the principal curvatures κ_1, κ_2; from the second fundamental form $L = \mathbf{S}_{uu}\cdot\mathbf{n}$, $M = \mathbf{S}_{uv}\cdot\mathbf{n}$, $N = \mathbf{S}_{vv}\cdot\mathbf{n}$ ($\mathbf{n} = \mathbf{S}_u \times \mathbf{S}_v / \|\mathbf{S}_u \times \mathbf{S}_v\|$), the Gaussian and mean curvatures are

$$K = \kappa_1\kappa_2 = \frac{LN - M^2}{EG - F^2}, \qquad H = \tfrac{1}{2}(\kappa_1 + \kappa_2) = \frac{EN - 2FM + GL}{2(EG - F^2)}, \tag{5.2}$$

and $\kappa_{1,2} = H \pm \sqrt{H^2 - K}$. The per-modality values in table 2 confirm the geometric intuition: developables have $K = 0$ (cylinder, cone, extruded), and the sphere is umbilic ($\kappa_1 = \kappa_2 = 1/r$). For the cone the single non-zero principal curvature is $\cos\alpha/(s\tan\alpha)$, growing without bound toward the apex; for the torus the two principal curvatures are $1/r$ (tube) and $\cos\phi/(R + r\cos\phi)$ (which changes sign across the inner/outer equator), so K is signed.

The surface sagitta argument mirrors (4.7) with $R = R_{\min} = 1/\kappa_{\max}$: a triangle edge of length ℓ on a surface of smallest principal radius $R_{\min}$ deviates from the surface by $\varepsilon \approx \ell^2/(8R_{\min})$, so the face sizing field is

$$h_{\text{face}}(\mathbf{x}) \;=\; \min\Big(h_{\max},\ \sqrt{8\,\varepsilon\,R_{\min}(\mathbf{x})}\Big), \qquad R_{\min} = 1/\kappa_{\max}. \tag{5.3}$$

5.3 Distance-faithful charts

A chart $\Phi:$ face $\to \mathbb{R}^2$ lets the planar CVT machinery act on the surface (figure 3). We want Φ as close to an *isometry* as possible so Euclidean 2D distance equals surface arc length.

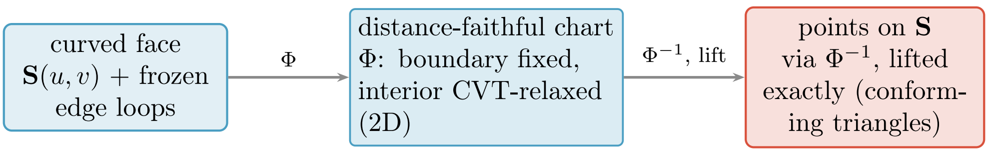

Figure 3: Surface meshing through a chart: relax in a (near-)isometric 2D parameterization with the frozen edge points as fixed boundary, then lift every chart point back onto the analytic surface. Exact on developables, bounded distortion on the sphere (equation (5.4)).

Table 2: Per-modality geometric quantities (all verified by computer algebra): the parameterization $\mathbf{S}(u, v)$, the first fundamental form E, F, G and area element $\mathrm{d}A$ (section 5.1), and the Gaussian curvature K and largest principal curvature $\kappa_{\max} = 1/R_{\min}$ that drive the deflection sizing (section 5.2). $\mathbf{a}$ is the unit axis, α the cone half-angle. All but the NURBS row are diagonal ($F = 0$); developables have $K = 0$; the sphere is umbilic.

Surface	$\mathbf{S}(u,v)$	E	F	G	$\mathrm{d}A$	K	$\kappa_{\max} = 1/R_{\min}$
Plane	$\mathbf{o} + u\mathbf{e}_1 + v\mathbf{e}_2$	1	0	1	$\mathrm{d}u\,\mathrm{d}v$	0	0
Cylinder (θ, h)	$r(\cos\theta, \sin\theta) + h\mathbf{a}$	r^2	0	1	$r\ \mathrm{d}u\,\mathrm{d}v$	0	$1/r$
Sphere (λ, φ)	$r(\cos\varphi\cos\lambda, \cos\varphi\sin\lambda, \sin\varphi)$	$r^2\cos^2\varphi$	0	r^2	$r^2\,\lvert\cos\varphi\rvert\ \mathrm{d}u\,\mathrm{d}v$	$1/r^2$	$1/r$
Cone (θ, s)	$s\tan\alpha(\cos\theta, \sin\theta) + s\mathbf{a}$	$s^2\tan^2\alpha$	0	$\sec^2\alpha$	$\lvert s\rvert\tan\alpha\sec\alpha\ \mathrm{d}u\,\mathrm{d}v$	0	$\dfrac{\cos\alpha}{s\tan\alpha}$
Torus (θ, ϕ)	$(R+r\cos\phi)(\cos\theta, \sin\theta)+r\sin\phi\,\mathbf{a}$	$(R+r\cos\phi)^2$	0	r^2	$r\,\lvert R+r\cos\phi\rvert\ \mathrm{d}u\,\mathrm{d}v$	$\dfrac{\cos\phi}{r(R+r\cos\phi)}$	$1/r$
Extruded (t, h)	$\mathbf{b} + P(t) + h\mathbf{a}$	$\lVert P'(t)\rVert^2$	0	1	$\lVert P'(t)\rVert\ \mathrm{d}u\,\mathrm{d}v$	0	$\dfrac{\lvert P' \times P''\rvert}{\lVert P'\rVert^3}$
NURBS (u, v)	tensor B-spline [9]	$\mathbf{S}_u\cdot\mathbf{S}_u$	$\mathbf{S}_u\cdot\mathbf{S}_v$	$\mathbf{S}_v\cdot\mathbf{S}_v$	$\sqrt{EG-F^2}$	$\dfrac{LN-M^2}{EG-F^2}$	$H+\sqrt{H^2-K}$

Developables (cylinder, cone, extruded): exact isometry. A surface with $K \equiv 0$ unrolls into the plane without distortion.

- *Cylinder*: $\Phi(\theta, h) = (r\theta,\ h)$. Then $E = G = 1, F = 0$ in the chart – exact.
- *Cone*: polar unroll $\Phi(\theta, s) = \big(\rho\cos\psi, \rho\sin\psi\big)$ with slant $\rho = s/\cos\alpha$ and sweep $\psi = \theta\sin\alpha$. The induced chart metric is $E = s^2\tan^2\alpha,\ F = 0,\ G = \sec^2\alpha$ – identical to the cone's first fundamental form (table 2), i.e. *exactly isometric*.
- *Extruded*: $\Phi(t, h) = (\sigma(t),\ h)$ with $\sigma(t) = \int^t \|P'\|$ the profile arc length; $E = G = 1, F = 0$ – exact.

Sphere: azimuthal-equidistant. The sphere ($K = 1/r^2 \neq 0$) admits no isometry to the plane. We use the azimuthal-equidistant chart about a cap centre: the chart radial coordinate is the exact geodesic distance $r\psi$ (ψ the geodesic colatitude), so radial distances are exact; the circumferential direction stretches by

$$\text{stretch}(\psi) = \frac{\psi}{\sin\psi} = 1 + \frac{\psi^2}{6} + \frac{7\psi^4}{360} + O(\psi^6), \tag{5.4}$$

which is ≤ 1.05 for caps up to $\psi \approx 32^\circ$ and stays bounded well past a hemisphere. The chart is a bijection on any cap that excludes the antipode; bijectivity is validated by a round-trip test $\text{dist}\big(\text{proj}_{\mathbf{S}}(\mathbf{p}),\ \Phi^{-1}\Phi(\mathbf{p})\big) < \text{tol}$.

Forward and inverse maps. Each chart provides Φ (surface point $\to$ chart, to place the fixed boundary) and Φ^{-1} (chart $\to$ surface, to lift relaxed interior points):

$$\begin{aligned}
\text{cylinder}: &\quad \Phi(\theta,h) = (r\theta, h), && \Phi^{-1}(x,y) = \mathbf{S}\big(x/r,\ y\big);\\
\text{cone}: &\quad \Phi(\theta,s) = \tfrac{s}{\cos\alpha}(\cos\psi, \sin\psi),\ \psi = \theta\sin\alpha, && \Phi^{-1}(x,y) = \mathbf{S}\Big(\tfrac{\operatorname{atan2}(y,x)}{\sin\alpha},\ \cos\alpha\sqrt{x^2+y^2}\Big);\\
\text{extruded}: &\quad \Phi(t,h) = (\sigma(t), h), && \Phi^{-1}(x,y) = \mathbf{S}\big(\sigma^{-1}(x),\ y\big);\\
\text{sphere}: &\quad \Phi = \log_{\mathbf{c}},\ \Phi^{-1} = \exp_{\mathbf{c}} && \text{(geodesic polar about } \mathbf{c}).
\end{aligned} \tag{5.5}$$

For the sphere, $\log_{\mathbf{c}}(\mathbf{p})$ returns the chart point at radius $r\psi$ ($\psi = \arccos(\hat{\mathbf{c}} \cdot \hat{\mathbf{p}})$ the geodesic colatitude) along the bearing of the tangent toward $\mathbf{p}$; $\exp_{\mathbf{c}}(\mathbf{u})$ walks the geodesic of length $\|\mathbf{u}\|$ from $\mathbf{c}$. The extruded σ^{-1} is the arc-length table inversion of section 4.1. Every lift lands *exactly* on the analytic surface, so the resulting boundary vertices satisfy the same on-surface guarantee as an explicit closest-point projection.

General NURBS / non-developable: metric-scaled or restricted CVT. For a NURBS patch we scale the parameters by the local metric ($\sqrt{E}, \sqrt{G}$) at the patch centroid (a first-order isometry) and, when a single chart distorts too much, fall back to a genuine restricted-Voronoi relaxation on the surface [5], [10]: the centroid (2.7) is computed from the restricted cells and the move is projected back onto $\mathbf{S}$ by closest-point.

Revolution faces and periodicity. A full revolution (cylinder/cone/torus barrel) is periodic in θ; its boundary loops are rim circles that do not close in the unrolled rectangle. Such a face is meshed by a structured (θ, v) grid whose rings reuse the *rim* longitudes (radial alignment, no seam) and whose row spacing follows h in arc length – the uniform grid is the on-surface CVT optimum of a developable. A sphere cap meeting an intersection circle is itself a revolution about the line of centres (the circle lies in a plane $\perp$ that axis), so the same latitude/longitude construction applies, with both incident caps deriving their longitude reference from the *shared* rim so their rings align.

5.4 Surface CVT and adaptivity

Algorithm 2: MeshFace($\mathbf{S}$, loops, h, ε)

Input: trimmed surface $\mathbf{S}$; frozen boundary loops; field h; tolerance ε
Output: conforming triangulation of the face (triangles + on-surface vertices)
build chart Φ (modality-driven); validate bijectivity;
$B \leftarrow \Phi$(frozen edge points of all loops) ; `// fixed boundary in chart`
scatter interior sites in Φ(face) at spacing h (greedy graded Poisson) inside the trim loops;
repeat
  for *a few iterations* **do**
    2D Delaunay of $B \cup \{\text{interior}\}$ (exact predicates);
    move each interior site by (2.7), $\rho = h^{-2}$, clamp inside the loops; keep B fixed;
  $e^\star \leftarrow$ max triangle deflection vs. $\mathbf{S}$ (sagitta, (5.3)) ; `// error-driven`
  $\ell^\star \leftarrow$ max triangle edge length (arc length) ; `//` $h_{\max}$`-driven`
  if $e^\star > \varepsilon$ ***or*** $\ell^\star > h_{\max}$ **then**
    insert a site at the offending triangle's circumcentre (in the chart);
until $e^\star \leq \varepsilon$ ***and*** $\ell^\star \leq h_{\max}$ ***and*** *max move* $< \tau_{\text{move}}$;
lift $\mathbf{p}_k = \mathbf{S}(\Phi^{-1}(\text{chart site}_k))$ – exact on surface;
return triangulation;

After Stage 2 every face carries a frozen, conforming triangulation; shared edges guarantee face-to-face conformity, and all faces run in parallel (Principle 1.4). The union of all face triangulations is a watertight surface mesh $\mathcal{S}$.

6 Stage 3 – Volume

6.1 Sizing field from the surface

The volume sizing field is grown from the completed surface: at each surface vertex the local feature size and the surface element size seed h_{vol}, which is then propagated through the interior and gradation-limited by (2.1). Crucially this is computed *after* the surface is final, so it already reflects curvature- and feature-driven refinement, including *local thickness*: where two boundary sheets are closer than $h_{\max}$ (a thin web, a blunt trailing edge, a slender bore wall) the field shrinks so the interior is resolved.

Local feature size and thickness. The local feature size $\text{lfs}(\mathbf{x})$ is the distance from $\mathbf{x} \in \mathcal{S}$ to the medial axis; sampling at spacing $\leq$ lfs guarantees a topologically faithful, well-shaped mesh [7]. We use the cheaper one-sided estimate

$$\text{thk}(\mathbf{x}) \;=\; \tfrac{1}{2}\, \text{dist}\big(\mathbf{x},\; \mathcal{S} \setminus \mathcal{S}_{\text{local}}(\mathbf{x})\big), \tag{6.1}$$

the half-distance along the inward normal ray from $\mathbf{x}$ to the first *opposite* surface sheet (excluding the patch $\mathcal{S}_{\text{local}}$ that $\mathbf{x}$ belongs to), and set the volume field to $h_{\text{vol}} = \min\big(h_{\text{face}}, \text{thk}/n_{\min}\big)$ so at least $n_{\min}$ elements span the thinnest gap. This is exactly what removes the straddling-tetrahedron failure mode at thin features before the CDT is built.

6.2 Interior seeding

Interior sites are thrown at the field density: a randomised, separation-guarded dart pattern (or a body-centred-cubic lattice graded by h [11]) with a per-point clearance from $\mathcal{S}$ of one local element, and a global budget $\int_\Omega h^{-3}\,\mathrm{d}V$ so a sharp feature does not inflate the count. Surface vertices are *not* re-seeded; they are the boundary of the volume point set.

6.3 3D CVT and quality-driven adaptivity

Interior sites relax by the 3D centroid rule (2.7) with $\rho = h^{-3}$; surface vertices stay fixed. A site whose centroid leaves the domain is mirrored back across the nearest boundary plane. After relaxation, sites are inserted where the local edge length exceeds $h_{\max}$ or where element quality (minimum dihedral angle, radius–edge ratio) is poor, and the relaxation is repeated.

6.4 Boundary-constrained tetrahedralization

This is the structural heart of the method. The volume is the *constrained Delaunay tetrahedralization* (CDT) of the interior sites *plus* the frozen surface mesh $\mathcal{S}$ as constraint facets [12], [13], [14]:

$$\text{Tetrahedralize}\big(\{\text{interior sites}\} \cup V(\mathcal{S})\big) \quad \text{s.t. every } \triangle \in \mathcal{S} \text{ is a union of tet faces.} \tag{6.2}$$

Because $\mathcal{S}$ is watertight and every constraint triangle is recovered as tetrahedron faces, each surface triangle has a tetrahedron exactly on each side; region labels are then assigned by flood fill across non-constraint faces starting from the region tag attached to each surface triangle's two sides. No tetrahedron straddles the boundary, so:

Per-region parallelism. Each material region is bounded by a disjoint subset of the frozen $\mathcal{S}$ (its outer walls and the interface facets it shares with neighbours, all read-only), so the regions' constrained tetrahedralizations are independent and run in parallel (Principle 1.4); a shared interface facet is meshed once on $\mathcal{S}$ and both regions tetrahedralize against the same fixed triangles, which keeps the multi-material mesh conforming across interfaces.

Proposition 6.1 (Watertightness)**.** *If $\mathcal{S}$ is a closed, non-self-intersecting, conforming surface mesh and the CDT recovers all of its facets, then for every constraint triangle the two incident tetrahedra carry the region labels prescribed by that triangle's front/back tags, and the boundary of each meshed region is exactly the corresponding subset of $\mathcal{S}$. In particular there are no protrusions into a void and no gaps.*

Boundary recovery proceeds in two classical stages [12], [13]: *edge recovery* forces each missing constraint edge by inserting Steiner points on it until it appears as a union of mesh edges; *facet recovery* locally re-tetrahedralizes the cavity straddling each constraint triangle so the triangle becomes a union of tetrahedron faces. Every Steiner point lies *on* $\mathcal{S}$ (never off it), so the boundary geometry is preserved exactly and proposition 6.1 still holds.

Quality measures and the cleanup pass. Element quality is reported as the minimum dihedral angle and the *radius–edge ratio* $\rho = R_{\text{circ}}/\ell_{\min}$; Delaunay refinement bounds ρ, but *slivers* (near-zero volume, dihedral $\to 0$, yet acceptable ρ) can survive and are the residual enemy. They are removed by a conformity-preserving post-pass: *(i)* smoothing each free vertex toward its quality-weighted optimum, with boundary vertices smoothed only *within their surface/edge carrier* (a tangential move; to first order it leaves each region's enclosed volume unchanged and keeps interface vertices exactly on their analytic carrier); *(ii)* topological flips and weighted-Delaunay *exudation* [15], [16], which assign vertex weights that "exude" slivers without moving any point. Both leave $\mathcal{S}$ and the region partition intact.

Algorithm 3: MESHVOLUME($\mathcal{S}, h$)

Input: watertight surface mesh $\mathcal{S}$ (with region tags); field h
Output: tetrahedra, per-tet region, boundary faces ($= \mathcal{S}$)
build h_{vol} from $\mathcal{S}$ (feature size, thickness), gradation-limit by (2.1);
seed interior sites at density h_{vol}^{-3}, clearance $\geq h$ from $\mathcal{S}$;
repeat
    CDT of (interior $\cup V(\mathcal{S})$) with $\mathcal{S}$ as constraints;
    for *a few iterations* **do**
        relax interior sites by (2.7), $\rho = h^{-3}$;
    insert sites where edge length $> h_{\max}$ ($h_{\max}$*-driven*) or element quality is poor (*quality-driven*);
until *size and quality targets met* ***and*** *max move* $< \tau_{\text{move}}$;
flood-fill region labels across non-constraint faces from surface tags;
quality post-pass (carrier-constrained smoothing, flips); drop slivers;
return tetrahedra, per-tet region, boundary faces;

7 The sizing field, end to end

The single field h threads all stages and is only ever *refined* downward in size:

1. **Edges** set h from curve deflection (4.9) and $h_{\max}$ / per-region overrides.
2. **Faces** take the min of the edge field and the surface deflection bound (5.3); the smoothed result drives the surface CVT.
3. **Volume** takes the min of the face field and the local-thickness / feature-size estimate, then gradation-limits (2.1) through the interior.

Because each stage only ever lowers h and then re-smooths, the field is monotone and Lipschitz, and the element-size contract (a documented $1.5\times$ max-edge bound, as in gmsh) holds globally.

8 Why this guarantees conformity

The earlier rapidmesh path built one *unconstrained* Delaunay over all points and recovered the boundary by labelling each tetrahedron with the region of its centroid. That recovers the boundary only *statistically*: where the surface sample is coarser than a feature, a tetrahedron straddles the true surface, and whole-tet labelling then produces a protrusion (centroid inside) or a gap (centroid outside). The present algorithm removes the failure mode at its root by (a) completing a watertight, error-controlled surface mesh *first*, and (b) making that mesh a *constraint* on the volume rather than a target to be rediscovered. Conformity is then a theorem (proposition 6.1), not a sampling-density gamble. The cost is the constrained Delaunay / facet recovery, which is exactly the machinery the method must keep; the variational (CVT) relaxation supplies the well-shaped interior that keeps recovery cheap and slivers rare [16], [17].

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